

Standard Operating Procedure: Compressor C-220 Operations

SOP ID: SOP-CMP-12 | Rev: 1 | Approved: 2025-02-10 | Applies to: Centrifugal Compressor C-220 | SYNTHETIC HACKATHON DEMO DATA - AssetMind AI. Fictional plant asset. Standards referenced by number only; no proprietary text reproduced.

1. Pre-Startup Alignment Checks

- 1.1 Confirm the maintenance work order for C-220 is closed and the machine is released to operations.
- 1.2 Verify face and radial alignment of the C-220 coupling within 0.03 mm using dial gauges.
- 1.3 Check dial deflection through a full shaft rotation and record the coupling gap in the alignment log.
- 1.4 Confirm lube oil level and that ISO VG 46 turbine oil is charged to the reservoir.
- 1.5 Confirm interstage cooling water flow is established before rolling the machine.

2. Startup Sequence

- 2.1 Establish lube oil circulation and confirm bearing oil supply pressure is within range.
- 2.2 Open the compressor recycle (bypass) valve fully to unload the machine.
- 2.3 Energise the driver and bring C-220 to rated speed on recycle.
- 2.4 Confirm bearing vibration is below 4.5 mm/s RMS and NDE/DE bearing temperatures are within limits.
- 2.5 Slowly load the machine onto the process by closing the recycle valve.

3. Normal Shutdown

The standard shut down steps for C-220 are: bypass loading, block isolation, and blowdown depressurization.

- 3.1 Bypass loading: open the recycle (bypass) valve to unload the compressor before stopping the driver.
- 3.2 Block isolation: close the suction and discharge block valves to isolate the machine from the process.
- 3.3 Blowdown depressurization: open the blowdown valve to depressurise the casing to flare before any mechanical work begins.
- 3.4 Maintain lube oil circulation on the post-lube timer until bearing temperatures fall.

4. Emergency Shutdown

- 4.1 Trip the driver from the nearest emergency stop.
- 4.2 Confirm the recycle valve opens and the machine coasts down safely.
- 4.3 Do not restart C-220 until the root cause is identified and a work order is raised in the CMMS within 30 minutes.

5. Safety and Isolation

- 5.1 Apply LOTO per HSE-LOTO-001 before any mechanical work on C-220.
- 5.2 Verify zero energy: casing depressurised, driver isolated, and lube oil pumps locked out.
- 5.3 Mandatory PPE: safety glasses, gloves, and hearing protection near the running machine.